SIADS 696 Milestone II Project Report

Spall Damage Prognostics from Gear Vibration Indicators

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Introduction

Gear teeth fail by spalling. Fatigue cracks under the tooth surface grow until pieces of the surface break away, starting as a small pit and growing run by run until the tooth breaks. The data come from a spur-gear test rig at the Gear Technology Center, where seven gears were run in blocks of about six hours until they spalled. After every block the rig was stopped, the gear was photographed and each of its 28 teeth was graded by eye on a four-point scale, and the organizers release vibration condition indicators (CIs), speed, torque, temperature and oil-debris counts for every second of running. The inspections are the expensive part. Each one means stopping the rig, pulling the gear and grading 28 teeth, while the indicators come from an accelerometer already on the rig. A model that reads damage from the indicators alone would let an operator inspect when the signal says to rather than on a fixed schedule.

Our project asks two questions. Can a gear's damage, as a severity class or a continuous health index, be predicted from the vibration indicators and context sensors alone, without inspection or teardown? And without any labels, how well does novelty detection against a gear's own healthy state alarm once the gear is no longer healthy? For Part A we trained multinomial logistic regression, a random forest and k-nearest neighbors to place each graded run end in one of three damage classes, and fit two label-based health indices, a non-negative ridge on the mean tooth damage and an index whose settings were tuned on the labels inside each fold. For Part B we fit a Mahalanobis distance with shrunk covariance, the squared prediction error of a PCA subspace and an Isolation Forest on healthy minutes only, and compared them against plain z-score aggregates with no model at all. Every result comes from the same leave-one-gear-out protocol, so classifiers and novelty detectors are compared on the same held-out gears, and the health index we submitted was chosen without labels, by the Coble and Hines fitness criteria, with the labels used only afterwards to check the choice. The headline finding is that the best honest supervised model, the index tuned inside each fold, is statistically tied with the best unsupervised one, and that the scarce resource is not labelled hours but labelled damage events.

Related Work

The condition indicators we use are decades old. Stewart (1977) proposed FM4, the kurtosis of what remains of the time-synchronous average once the mesh harmonics are removed; Zakrajsek, Townsend and Decker (1993) tested FM0, FM4 and a new indicator, NA4, on spur gears run to failure by surface pitting; and Decker (2002) did the same kind of study for tooth cracks. In each of these the gear is its own reference and the question is whether one indicator moves before the gear fails. We have seven gears, so we can hold one out and ask whether an indicator that worked on the others also works on it. That is how we found that every rig stop moves the global indicators further than an early spall does, and that only the four residual-signal indicators (FM4, M6A, M8A, NA4) tell the two apart. Coble and Hines (2009) defined three properties of a good prognostic parameter, monotonicity, prognosability and trendability, and used a genetic algorithm to search for feature weights that maximize them. We keep the three properties and drop the search. The candidate health indices were fixed before we saw any labels, the one with the best fitness was submitted, and the labels were used only afterwards to check it. Randall (2011) describes the running-in period, when vibration is high before it settles, and our per-gear healthy-window standardization and the Part A distance-from-running-minimum features follow from that. None of this prior work compares supervised classifiers, a label-tuned index and label-free novelty detectors on the same held-out gears under one protocol, which is what this report adds. This project uses a new dataset and is not an extension of any team member's Milestone I project.

Data Source, Scope and Preprocessing

Source and scope

The dataset originates from the *Surface Damage Progression* track of the PHM Society 2026 Data Challenge, acquired directly from the organizers' repositories. Each experiment tracks an individual gear over its operational lifespan. Table 1 outlines the scope and characteristics of the provided data across all test units. For the purposes of this report, our analysis focuses exclusively on the four training gears, as they are the only units accompanied by inspection labels. The remaining gears, including validation Gear G as well as test Gears D and E, are used solely for evaluation and are strictly excluded from all model fitting and internal validation procedures.

Link to source: https://data.phmsociety.org/phm-north-america-2026-conference-data-challenge/

Table 1. The seven gears. *Hours represent test hours of logged running, rows are one-second records, files are per-minute HDF5 files, and photos are tooth photographs. Spalled teeth at the end counts teeth whose final condition grade reached 3 for spalling or higher.*

Gear	Split	Runs	Hours	1-s rows	HDF5 files	Photos	Photo inspections	Spalled teeth at end	Torque (ft-lbf)	Oil temp. (°F)
Exp-A	*train*	5	30.5	110,205	1,833	455	7	3	120.0	129.6
Exp-B	*train*	7	39.4	142,082	2,375	576	9	2	100.0	120.2
Exp-C	*train*	12	73.9	267,021	4,460	774	13	1	99.9	120.9
Exp-F	*train*	8	43.8	158,527	2,910	280	10	5	100.0	120.0
Exp-G	*validation*	16	98.7	356,514	5,889	0	–	–	100.0	119.3
Exp-D	*test*	9	35.9	129,747	2,162	0	–	–	100.0	117.9
Exp-E	*test*	12	73.9	266,999	4,456	0	–	–	100.0	118.0

Across all seven gears, the dataset recorded 22,763 logged minutes of running time, accompanied by 2,085 photographs and 1,092 individual tooth inspections.

Signals

Every per-minute HDF5 file contains roughly 60 one-second values for each channel, which required processing during extraction. Our pre-processing pipeline extracts the PHM files and converts them into a single parquet table for each split, as described in Appendix B. We group these channels into three distinct categories.

- Condition Indicators: These are scalar statistics calculated by the organizers using the high-frequency accelerometer signal, mostly derived from the time-synchronous average of a single shaft revolution. Twelve of these indicators offer a clear damage interpretation without relying on external rig runs, making them the primary inputs for every model. Four specific indicators, including FM4, M6A, M8A, and NA4, are styled as tooth CIs because they capture localized tooth damage by measuring the impulsiveness of the residual signal after removing regular mesh harmonics. The remaining eight indicators respond to overall vibration levels.

- Context: This includes shaft speed, torque, oil temperature, and the RMS from two accelerometers. Speed and torque define the steady-state filter, whereas temperature and torque are regressed out from each condition indicator.
- Oil Debris: These consist of particle counts across eight size bins collected from two debris monitors, recorded at one-minute intervals. The six largest ferrous bins from the DM4500 monitor form our debris baseline.

Labels

Following each run, the gear was photographed and each individual tooth received a grade on the organizers' condition scale, where 1 represents healthy, 2 indicates spall onset, 3 represents spalling, and 4 indicates severe spalling, with fractional grades permitted. We derive two run-level metrics from these individual tooth grades.

- Mean tooth damage: This value equals the average over graded teeth of the score minus one. It serves as our primary target T1, taking a value of 0 for a healthy gear and reaching 3 if every tooth exhibits severe spalling. Every error reported in this paper divides this value by 3 to bound it within the 0 to 1 range.
- Worst tooth: This represents the maximum grade observed. A run end qualifies as a spall event when the worst tooth reaches a grade of 2 for onset. Across the 32 training run ends, we identify 11 spall run ends, comprising one on Exp-A, four on Exp-B, one on Exp-C, and five on Exp-F.

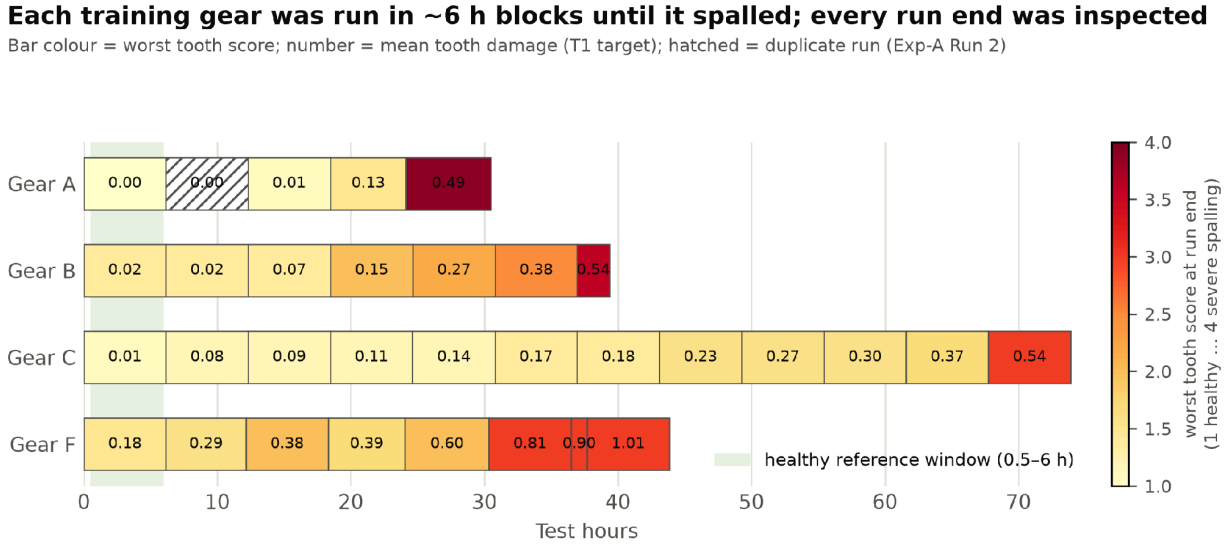


Figure 1. Run-to-failure timeline of the four training gears. *Each bar is one ~6 hour run. Its color is the worst tooth grade at the run end and the printed number is the mean tooth damage. The hatched bar is Exp-A Run 2, a byte-for-byte duplicate of Run 1 in the released data, which we blank. The green band is the healthy reference window (0.5–6 h) that every model is standardised against.*

Preprocessing pipeline

Every model in Parts A and B consumes the same table of condition-corrected steady minutes. This table is produced and applied inside each cross-validation fold through the following sequential steps.

1. Steady-state filter: The pipeline retains one-second rows operating at a nominal speed of 1400 plus or minus 20 rpm with torque within five percent of the median for that gear. Each minute is then reduced to the median of its constituent rows.
2. Log and condition correction: The pipeline takes the logarithm of the absolute value of each condition indicator and regresses out oil temperature and torque for each gear. Oil temperature climbs by

about 35 degrees Fahrenheit during the first hour of every run, as shown on the right side of Figure A4. Without this regression step, every condition indicator reflects that initial warm-up climb.

3. Per-gear robust standardization: Each condition indicator is centered and scaled using the median and median absolute deviation of the healthy window for that gear, which covers the period from 30 minutes to 6 hours inside Run 1. The first half hour represents the initial run-in and is skipped. Through this transformation, every feature measures how many healthy standard deviations a gear has moved from its baseline state.
4. Run-end value: For each run, the pipeline extracts the 90th percentile of the per-minute score calculated over the final 60 steady minutes. Because post-spall vibration impulses are short-lived, taking a simple mean or median over a long tail would obscure these critical signals.
5. Monotone step: Because physical spalls do not heal, the sequence of run-end values is constrained to be non-decreasing for each gear using isotonic regression. This processed sequence serves as the official score evaluated by the challenge metric. Additionally, a causal running maximum of the raw run-end values is preserved to trigger real-time alarms and evaluate lead times without introducing information from subsequent runs.

Known data issues

- Run 2 of Exp-A is a byte-for-byte duplicate of Run 1 in the source files, so its minute-level records are removed from processing. Its label is retained for timeline alignment, but Exp-A contributes four scored run ends with sensor data to ridge model training and five run ends to evaluation, where the duplicate run end receives the carried-forward value.
- Operating Conditions: Exp-A operated at 120 ft-lbf of torque and approximately 130 degrees Fahrenheit, whereas all other gears operated at 100 ft-lbf and approximately 120 degrees Fahrenheit. Per-gear standardization absorbs these baseline operational shifts, but it cannot predict how a model trained at 100 ft-lbf will perform under a 120 ft-lbf load.
- Camera Configurations: Exp-C utilized two distinct camera setups during inspection without detailed documentation explaining the methodological differences between them.
- Short Operational Runs: Run 7 of Exp-B lasted 2.4 hours, Run 7 of Exp-F lasted 1.2 hours, and Exp-D contained four abbreviated runs. Any run shorter than the standard 60-minute evaluation tail is scored using all available steady minutes for that run.
- Inspection Label Noise: On Exp-F, the worst-tooth score progresses non-monotonically from 2.0 to 1.7 and back to 2.0 across Runs 3 through 5, which represents a physical impossibility. To account for this uncertainty, inspection scores are treated as noisy, and performance rankings are reported alongside secondary evaluations using only the final two, most thoroughly verified inspections per gear.

Supervised Learning (Part A)

Data and Feature Representation

The target variable, gt_condition, was mapped into three ordinal damage classes: 1 for values from 1 to 1.5, 2 for values above 1.5 up to 2.5, and 3 for values at or above 2.5. This mapping was applied consistently across all four experiments used in training and testing.

The initial candidate feature set included ci_fleet_wtmade_norm_dist_from_min_so_far and ctx_accel1_rms_dist_from_min_so_far, each expressed as a distance from each unit's own running minimum, along with PCA_dist_from_min_so_far, derived from a PCA fit on three normalized condition indicators (ci_fleet_wtmade_norm, ci_fm4star_norm, ci_m6astar_norm). This construction is intended to capture monotonic degradation relative to the healthiest state observed so far for a given unit.

These features were expressed as a distance from the running minimum, rather than as raw normalized values, due to a suspected wear-in effect: rotating machinery can show elevated vibration readings during an initial running-in period before stabilizing at a lower baseline and later rising as damage develops

(Randall, 2011). Anchoring to the running minimum rather than the first observed value is intended to let this early elevation settle before the feature tracks subsequent degradation.

Missing values were handled with median imputation and all features were standardized, both implemented inside a scikit-learn pipeline refit within each training fold so no held-out experiment's data influenced its own preprocessing.

Methodology

Model evaluation used a leave one experiment out design across the four available experiments. In each of four folds, one experiment was held out as the test set and the remaining three were pooled for training. This experiment level split is intended to keep each held out experiment's data separate from training, since the experiment is the primary unit of variability available in this dataset.

Three model families were evaluated under this protocol. 1) multinomial logistic regression, 2) random forest, and 3) k-nearest-neighbors. All three used identical preprocessing and the same four fold split. Feature selection was performed before hyperparameter tuning, with the resulting feature set then held fixed while model settings were explored.

Feature Engineering and Optimization

Before tuning hyperparameters, the feature set was systematically searched. The candidate list was expanded beyond the original condition indicators to include normalized condition indicators, their distance-from-running-minimum transforms, and the PCA output in raw and transformed form. Every combination was run through the same four fold protocol using each model's default settings, scored by fold averaged macro-F1, accuracy, and weighted-F1.

The strongest overall combination paired **ci_fleet_wtmade_norm** and **ctx_accel1_rms_dist_from_min_so_far** with a random forest, reaching mean macro-F1 0.7431 and accuracy 0.7827. The same pairing was also best for KNN (macro-F1 0.6931, accuracy 0.7262). Logistic regression reached its best macro-F1 of 0.6796 with this pairing, the original three-feature set, and one

Table A1 — Selected feature pair vs. next-best alternative (default hyperparameters, 4-fold LOEO)

Feature combination	Model	Accuracy	Macro-F1	Macro-F1 SD
ci_fleet_wtmade_norm + ctx_accel1_rms_dist_from_min_so_far (selected)	RF	0.7827	0.7431	0.0814
ci_fleet_wtmade_norm + ctx_accel1_rms_dist_from_min_so_far	KNN	0.7262	0.6931	0.1022
ci_fleet_wtmade_norm + ctx_accel1_rms_dist_from_min_so_far	LR	0.7202	0.6796	0.0712
m6astar + PCA (next-best RF alternative)	RF	0.7768	0.7242	0.2571

Feature Ablation

A feature by feature marginal value analysis, comparing each model's average macro-F1 with and without a given feature across all combinations containing it, found the largest positive contributions from ctx_accel1_rms_dist_from_min_so_far (logistic regression), ci_fleet_wtmade_norm (all three models), and PCA_dist_from_min_so_far (random forest). The raw, untransformed ctx_accel1_rms feature had a

negative marginal value for all three models. Performance was consistently better with it excluded in favor of its distance from minimum counterpart. This pattern is consistent with, though does not confirm, the wear in thought process described above.

Table A2 - Marginal Value of each feature

Feature	Model	Avg_F1_With_Feature	Avg_F1_Without_Feature	Delta
ci_fleet_wtmade_norm_dist_from_min_so_far	LR	0.499008	0.344384	0.154624
ci_fleet_wtmade_norm	LR	0.479563	0.349384	0.13018
ctx_accel1_rms_dist_from_min_so_far	LR	0.462307	0.353821	0.108486
ci_fleet_wtmade_norm_dist_from_min_so_far	KNN	0.52123	0.420915	0.100315

Hyperparameter Exploration

With feature selection treated as settled, an exploratory sweep was conducted over several settings per model family, holding features fixed. The number of neighbors for KNN (3, 5, 7), inverse regularization strength for logistic regression (0.1, 1.0, 10.0), and number of trees/max depth for random forest (300 or 800 trees; depth capped at 5 or unlimited). Each configuration was run through the same four-fold protocol, with accuracy and macro-F1 averaged across folds. The sweep was run on the final two-feature representation; Table A3 reports the best setting per model.

Table A3 — Tuned results, two-feature representation (4-fold LOEO)

Model	Setting	Accuracy (mean ± SD)	Macro-F1 (mean ± SD)
RF	800 trees, max_depth=None	0.7827 ± 0.1772	0.7431 ± 0.0814
KNN	k=7	0.7619 ± 0.2103	0.7232 ± 0.1058
LR	C=1.0	0.7202 ± 0.2079	0.6796 ± 0.0712

Based on this exploration, the configurations carried forward are logistic regression (C=1.0), random forest (800 trees, depth=None), and KNN (k=7).

Results

With the selected feature pair and the tuned settings, the random forest has the highest fold-averaged macro-F1 (0.7431 ± 0.0814) and accuracy (0.7827 ± 0.1772), followed by KNN (0.7232 ± 0.1058) and logistic regression (0.6796 ± 0.0712), which has the narrowest spread across folds. Macro-F1 is the headline metric because the 29 test records split 13/11/5 across the three classes and it weights the rare severe class equally with the others; accuracy, macro precision and recall, and weighted-F1 are in Table A4.

Table A4 - Model performance across 4 experiments (LOEO)

Model	Accuracy	Macro_F1	Macro_Precision	Macro_Recall	Weighted_F1
KNN	0.761905	0.723214	0.790278	0.748016	0.737847
LogReg_multinomial	0.720238	0.679563	0.741667	0.720238	0.700149
RandomForest	0.782738	0.743081	0.793981	0.759921	0.769867

Logistic regression performed perfectly on EXP_A, correctly classifying all three class-1 samples and the single class-3 sample. Its performance dropped on EXP_C, where it correctly identified all four class-1 samples but only one of seven class-2 samples, routing the rest into class 1. The random forest showed a similar pattern on EXP_C, correctly identifying all four class-1 samples but none of the seven class-2 samples in this configuration. K-nearest-neighbors was more variable, performing its worst on EXP_A and its best on EXP_C and EXP_F.

Pooling the confusion matrices across all four folds and 29 test records gives a consistent picture. For every model, the bulk of the classification errors occur at the boundary between class 1 and class 2, while confusion between class 1 and class 3, or between class 2 and class 3, is rare. The class-2 difficulty is strongest in EXP_C specifically, where all three models classify the large majority of true class-2 samples into class 1.

Table A5- Aggregated confusion matrix (Summed across al LOEO folds)

	LogReg_Multinomial			Random Forest			KNN		
	Pred Class 1	**Pred Class 2**	**Pred Class 3**	**Pred Class 1**	**Pred Class 2**	**Pred Class 3**	**Pred Class 1**	**Pred Class 2**	**Pred Class 3**
True Class 1	11	2	0	12	1	0	12	1	0
True Class 2	7	4	0	6	5	0	7	4	0
True Class 3	1	0	4	1	0	4	1	0	4

Discussion

These results suggest logistic regression's linear boundary generalizes consistently across folds regardless of feature subset, while random forest and KNN are more feature sensitive and benefited from the simpler two feature representation. Random forest's larger fold to fold variance throughout both the feature search and the sweep further suggests greater sensitivity to which experiment is held out.

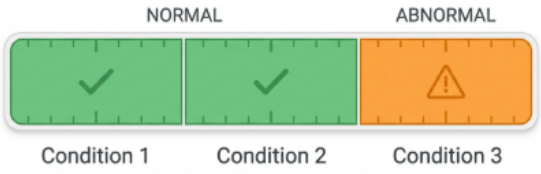
The recurring difficulty distinguishing class 1 from class 2, especially in EXP_C, suggests these two classes sit closer together in feature space than class 3 does to either one . The feature engineering results indicate the class 1/2 boundary can be sharpened somewhat, particularly for random forest, by combining the fleet-wear indicator with the accelerometer RMS distance feature rather than relying on the PCA-based feature.

From Three-Class Results to a Binary Framing

These three class results offer a partial motivation for also examining a simplified two class framing. A normal category combining the two milder damage levels, and an abnormal category for the most severe damage. As noted above, class 3 was comparatively easier to separate in most folds, while the class 1/2 boundary remained the dominant source of confusion. This is a rationale grounded in the observed results, not documented original reasoning for the binary study.

Binary Classification Study

A binary version of the labeling scheme was explored. gt_condition at or below the class 1,2 boundary labeled as normal, values above 2.75 labeled as abnormal.



This produced a strongly imbalanced split- EXP_A 3 normal/1 abnormal, EXP_B 6/1, EXP_C 11/1, EXP_F 5/1 25 normal and 4 abnormal total. Since each fold contains only a single abnormal example, any accuracy or recall computed for the abnormal class is estimated from one observation per fold.

Table A6 - Binary Classifier performance

model	accuracy	recall_abnormal	precision_abnormal	f1_abnormal	roc_auc	pr_auc
RandomForest	1	1	1	1	1	1
ThresholdBaseline	0.958	1	0.875	0.917	1	1

These results should be read as a promising preliminary direction rather than a validated measure of generalization to severe spall detection. More abnormal examples would be needed before drawing firm conclusions.

Conclusion

A systematic feature search found that pairing the fleet-wear indicator with the accelerometer RMS distance from min feature and a random forest model produced the strongest result to date, with mean macro-F1 0.7431 and mean accuracy 0.7827 across four leave-one-experiment-out folds, with the clearest gains in the two folds where the broader three-feature set struggled most with the class 1/2 boundary. This reinforced the value of expressing the accelerometer feature relative to each unit's running minimum, caused by a suspected wear-in effect (Randall, 2011). The dominant source of error throughout was confusion between the two milder damage classes, concentrated especially in EXP_C, while the most severe class was less often confused with the others in most folds. These findings motivate a simplified normal vs abnormal framing, explored separately with encouraging but statistically limited preliminary results given the small number of abnormal examples available.

Failure Analysis

1. Class 1/2 confusion is the dominant error mode

Across all four folds, the overwhelming majority of errors fall between classes 1 and 2 rather than involving class 3. This suggests classes 1 and 2 are harder to separate in the current feature representation than either is from class 3. Proposed improvement: The binary classification discussed earlier could be an approach that would solve this issue and have the model detect when damage has occurred to the gears and inspection is required.

2. Class 3 (severe spalling) errors are rare

Since class 3 usually only occurs on the last run of the experiment there are fewer class 3 examples compared to the other 2 classes. Proposed solution:The experiments could have been continued for additional runs once spalling was detected, this would give more cond 3 labels and corresponding signal data.

3. EXP-A more difficult to make good predictions for

In the beginning stages EXP-A proved to be difficult to get good predictions for across all models tested. This was due to an unusually high vibration signal on run 1 which dropped for subsequent runs. Solution: A new feature was created which calculated the distance from min so far which produced a more monotonic signal.

Tradeoffs

The random forest has the best macro-F1, 0.74 against 0.68 for logistic regression, but only with one feature pair; the next-best pair drops to 0.72, while logistic regression scores 0.68 on three different feature sets. Every model sends most true class-2 run ends to class 1 (recall 36–45%, Table A5). Moving the boundary would catch more onsets but trigger more empty inspections, and on a gearbox the missed onset costs more. The binary framing separates the severe class but leaves one abnormal record per fold. Fold-averaged accuracy (0.78 for the forest) beats pooled (21/29, 0.72) because EXP_A's four perfect records count as a whole fold.

Unsupervised Learning (Part B)

Motivation

Damage labels are scarce and expensive because each one requires stopping the rig, removing the gear, and inspecting all 28 teeth. A monitoring system on a new gear operates without any labels at all. We therefore framed Part B as novelty detection relative to a healthy reference. Each model is fit only on minutes in which a gear is known to be healthy, and damage is scored as the distance of later minutes from that healthy state. We use two facts about the test design rather than labels, which are that every gear starts healthy and every training gear was run until it spalled. The per tooth condition labels where 1 represents healthy and 4 represents severe spalling are used only to evaluate the finished models, and they are never used to fit models or to choose settings.

Data Source

Every model consumes the condition corrected steady minutes described in the Data section, and every stage of that pipeline is applied inside the leave one gear out folds. After the steady state filter, 10586 minutes remain across the four training gears, with 1412 on Exp A, 2300 on Exp B, 4324 on Exp C, and 2550 on Exp F. Each gear condition indicators are referenced to its own healthy window covering the first 30 minutes to 6 hours of Run 1, so every feature reads as the number of healthy standard deviations by which the gear has moved away from its own normal. The novelty model itself is fit only on the pooled healthy window minutes of the training gears, totalling 1320 baseline minutes, and then scores every minute of the held out gear. A health index for a run is the 90th percentile of its minute scores over the last 60 steady minutes, and the sequence of run end values is made non decreasing within each gear by isotonic regression.

Feature Representations

We compare two feature sets motivated by gear diagnostics theory and by our exploratory data analysis.

- All CI with 12 features: FM4, M6A, M8A, NA4, energy ratio, TSA kurtosis, TSA RMS, ALR, FM0, TSA crest factor, as well as the kurtosis and RMS of the energy operator.
- Tooth CI with 4 features: FM4, M6A, M8A, NA4. These indicators are designed to respond to localized tooth damage such as impulsiveness in the residual signal once the regular mesh harmonics are removed, as noted by Stewart in 1977, Zakrajsek et al. in 1993, and Decker in 2002. In exploratory data analysis, they stayed near zero on healthy runs, while the energy and global indicators shifted by tens of sigma after every rig restart.

Model Families

- We compare three mechanisms, each fit only on healthy minutes. The Gaussian density model uses a Ledoit–Wolf shrunk covariance and scores each minute by its Mahalanobis distance over all 12 CIs

or over the four tooth CIs, counting only upward departures. Shrinkage keeps the estimate stable despite the strongly correlated CIs. The linear subspace model applies PCA retaining 95% of variance and scores each minute by its squared prediction error (SPE). SPE flags patterns that healthy data never showed, even when each individual CI is within range. Isolation Forest uses 200 trees and scores points as anomalous when random splits isolate them quickly. It makes no distributional or linearity assumptions, which suits the heavy tails and multimodality of the CIs.

- To separate the effect of the model from that of the representation, three z-score aggregates on the same per-gear reference serve as a no-model control. The first, zpos, is the RMS of the positive z-scores across all 12 CIs. The second, zmax, is the largest absolute z-score across all 12 CIs. The third, tooth, is the mean positive z-score across the four tooth CIs and is our submitted HI.
- Tuning on labels would make the method semi-supervised, so every setting was fixed in advance. The healthy window spans 0.5–6 h, the longest window that lies inside Run 1 of every gear. The run-end value is the 90th percentile of the last 60 minutes, because post-spall impulses are short-lived and a median would wash them out. PCA keeps 95% of variance, and Ledoit–Wolf shrinkage is estimated analytically. Isolation Forest uses the scikit-learn defaults with 200 trees and seed 696 (Pedregosa et al., 2011). The alarm threshold is the 95th percentile of each method's score on the fit gears' healthy minutes, so about 5% of healthy minutes alarm by construction.
- To select among HIs without labels, we combined three criteria from Coble and Hines (2009) into one fitness score (Table B2). Monotonicity is the size-weighted share of rises minus falls. Trendability is the lowest pairwise correlation between gears' trajectories. Prognosability is the spread of end-of-life values relative to how far each gear's HI rose.

Evaluation – Protocol and Metrics

- Unit of evaluation. The gear, not the minute. Minutes from one gear share its rig, load and damage path, so a minute-level split would leak gear identity. Leave-one-gear-out CV refits the healthy model on three gears and scores the fourth, giving 32 scored run ends.
- Primary metric. Pooled monotone MSE on T1. The challenge scores an HI only up to a monotone rescaling. We therefore fit one isotonic map from HI to T1, shared across all four gears, and report the MSE after applying it. Sharing the map penalizes an HI that ranks runs well within each gear but sits at a different level on each. Per-gear calibration would hide that failure.
- Secondary metrics. Spearman ρ and per-gear error with its spread. Detection metrics come from a causal running-maximum alarm: spalls caught and missed, false alarms before onset, and lead time. Label-free fitness is the only metric usable without labels.
- Significance. With four gears, the exact sign-flip test against hours cannot go below p = 1/16. We therefore report per-gear wins out of four and treat bootstrap ranges as descriptive.
- Baselines. Elapsed hours, and debris rate (the large-particle rate relative to the first run), an independent non-vibration signal.

Overall results

Table B1. *Leave-one-gear-out results by method family (lower T1 MSE is better). Best per family in bold. The gear-error range is a descriptive 95 % bootstrap over the four held-out gears.*

Family	Method (features)	T1 pooled MSE	Spearman ρ	Gear error, mean [range]	Beats hours (k/4)	Spall caught / missed	False alarms	Median lead (h)
Gaussian density	**Mahalanobis (tooth CIs)**	**0.00175**	**0.875**	0.0016 [0.0004, 0.0032]	**4**	11 / 0	16	12.3
Gaussian density	Mahalanobis (all CIs)	0.00412	0.613	0.0049 [0.0018, 0.0081]	2	11 / 0	20	21.4

Linear subspace	**PCA SPE (all CIs)**	**0.00419**	0.605	0.0051 [0.0017, 0.0085]	2	11 / 0	21	21.4
Tree ensemble	**Isolation Forest (all CIs)**	**0.00318**	0.679	0.0037 [0.0018, 0.0056]	2	11 / 0	18	15.3
No-model control	Tooth z-score mean (submitted)	0.00179	0.863	0.0016 [0.0004, 0.0032]	4	10 / 1	13	3.1
No-model control	Positive all-CI z RMS	0.00261	0.718	0.0026 [0.0010, 0.0041]	3	11 / 0	15	9.2
Baseline	Elapsed hours	0.00432	0.657	0.0040 [0.0008, 0.0084]	–	11 / 0	18	15.3
Baseline	Debris rate	0.00237	0.662	0.0027 [0.0011, 0.0043]	2	6 / 5	11	–5.8

Table B2. *Label-free selection criteria (no labels used; higher fitness is better).*

HI	Monotonicity	Trendability	Prognosability	Run-1 false-alarm rate	Fitness
tooth (z-score mean)	**0.859**	**0.834**	0.631	**0.026**	**0.749**
tooth_maha	0.791	0.798	0.514	0.034	0.667
zpos	0.829	0.446	0.644	0.089	0.551
zmax	0.745	0.301	0.677	0.090	0.484
maha (all CIs)	0.802	0.317	0.657	0.176	0.415
spe (all CIs)	0.670	0.094	0.547	0.311	0.126

The three main findings follow.

1. Feature representation matters more than model family. All-CI models scored 0.0032–0.0042, barely better than elapsed hours (0.0043). Restricting the four tooth CIs cut error by 58% for Mahalanobis (0.00412 → 0.00175) and by 39% for Isolation Forest (0.00318 → 0.00194). Figure B1 shows why. All-CI scores jump 10–20× above threshold at each gear's first restart and stay there, even on Gear A's healthy Runs 3–4 (mean damage ≤ 0.13). The tooth-CI model rises only near spall onset. Figure B2 places those healthy runs far from the healthy cloud (PC2 ≈ −65). This is a reassembly shift that SPE cannot distinguish from damage.

2. Model families agree with each other more than with damage. Mahalanobis, PCA SPE and max z-score share near-identical per-gear errors (A and F ≈ 0.008, C ≈ 0.0002; Figure B3). They also share lead times and similar trendability (0.30–0.32; SPE 0.09). Once the input carries the restart shift, the mechanism hardly matters. Isolation Forest is the best all-CI model, plausibly because its axis-aligned splits lean less on correlated global CIs. It is also the least stable (see Sensitivity analysis).

3. The label-free rule picks a top model. Table B2 ranks tooth first (fitness 0.749) without labels. On T1 it ties tooth_maha (0.00179 vs 0.00175, well within the cross-gear spread). Both beat elapsed hours on all

four gears, the best result four gears allow (sign-flip p = 0.0625). An HI chosen from unlabelled data should therefore be near-best in deployment. Debris is competitive (0.0024) but misses 5 of 11 spalls because particle counts lag vibration.

Detection

All the vibration HIs caught at least 10 of the 11 spall run ends. The 13 to 21 false alarms listed in Table B1 overstate the problem. They represent flags on run ends before any tooth reached the onset score of 2, but on Gears B, C, and F, inspected damage was already rising by then, as seen in the bottom row of Figure B5. Many of these flags serve as early warnings of wear that has not yet reached the onset grade. The tooth-CI model gives no warning on Gear A, where the spall appeared within the final run, and warns 6 hours before onset on Gear F.

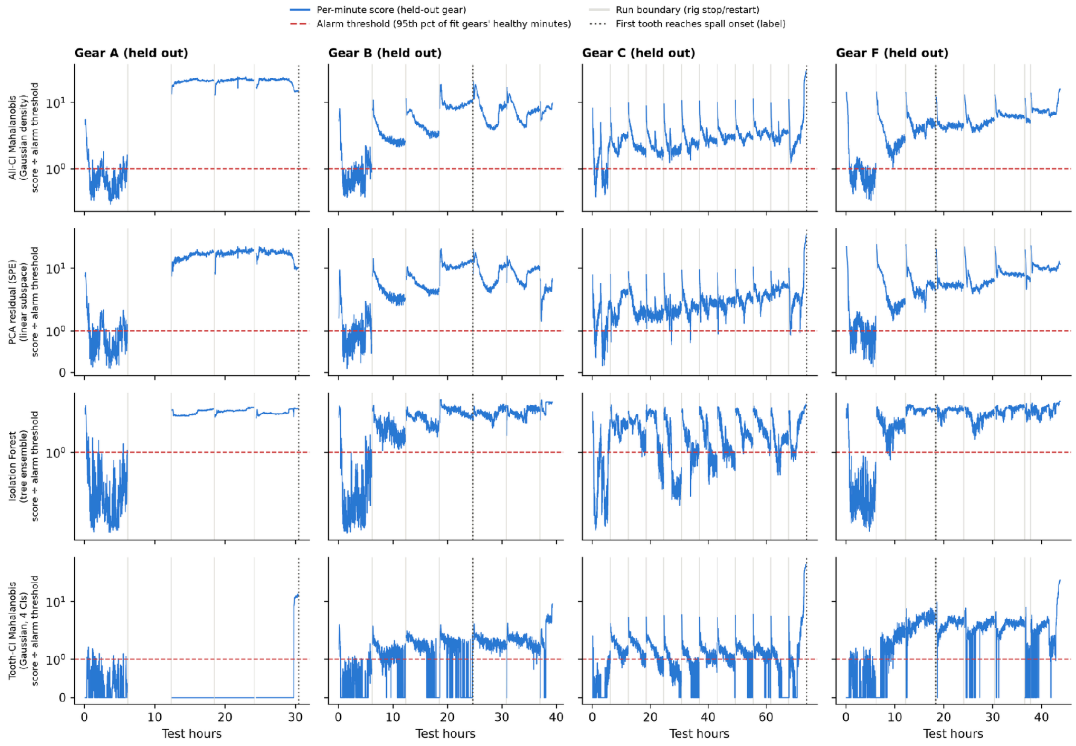


Figure B1. *Per-minute held-out scores for each method family (rows) and gear (columns), divided by the fold's alarm threshold. The y-axis is linear below 1 and log above. Grey verticals mark rig stops. The gap in Gear A is the duplicated Run 2. The all-CI models (rows 1–3) step up at the first restart and stay high on runs the labels call healthy. The tooth-CI Mahalanobis model (row 4) stays near or below threshold until spalling.*

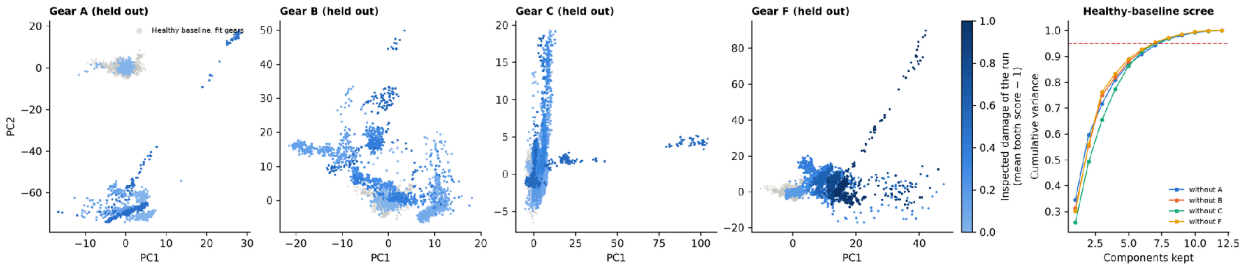


Figure B2. *PCA health space. Left four panels: the held-out gear's minutes projected onto the first two principal components of the other gears' healthy baseline (grey), coloured by the inspected damage of their run. Right: scree of the healthy baseline per fold; the dashed line marks the 95 % retention cut-off (7–8 components). On Gear F, damage grows along a consistent direction. On*

Gear A, the healthy post-restart runs (light) sit in the same far cluster as the spalled final run (dark), so distance from the healthy subspace alone cannot separate them.

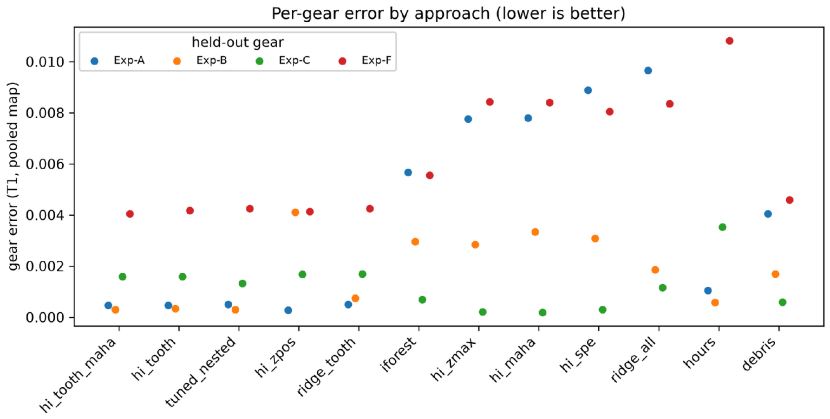


Figure B3. *Per-gear error (T1) for every approach, including the Part A models. Each dot is one held-out gear. The all-CI models (hi_maha, hi_spe, hi_zmax, iforest) and the all-CI ridge (ridge_all) fail on the same two gears, A and F. Exp-F is the hardest gear for the tooth-CI models because its damage is spread over five spalled teeth while the CIs track the worst tooth.*

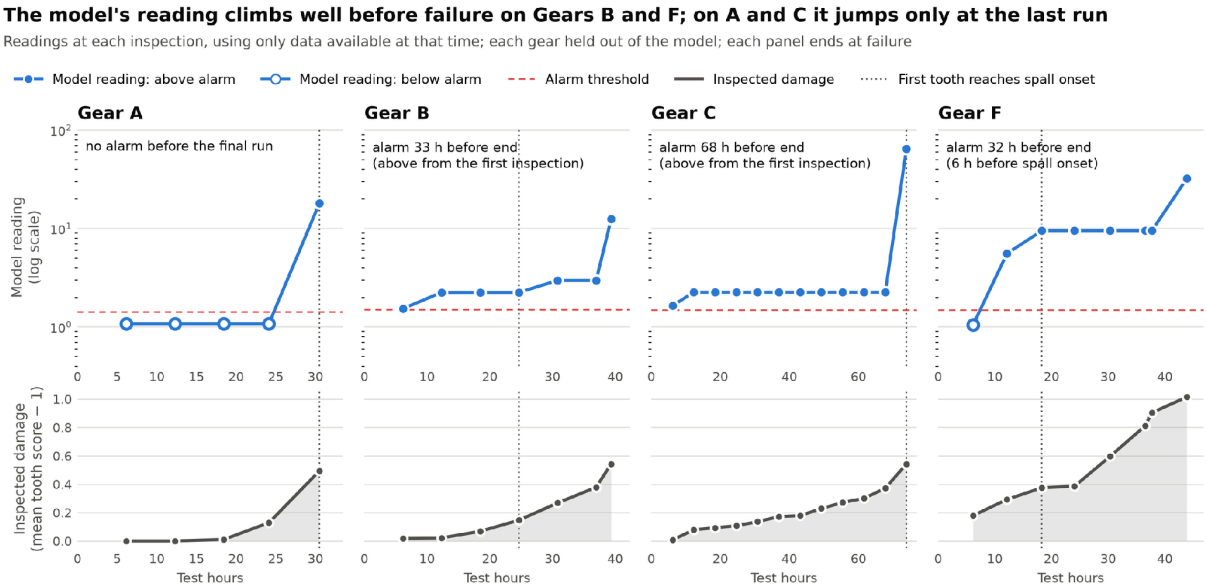


Figure B5. *Causal early warning of the tooth-CI Mahalanobis HI. Top: the run-end reading (log scale) and the alarm threshold. Bottom: inspected damage.*

Sensitivity analysis

The result is insensitive to the choices we fixed in advance. The baseline window of 1.5 to 6 hours yields a T1 of 0.00174 to 0.00179. The tail length of 10 to 240 minutes yields a T1 of 0.00179 to 0.00184.

The one setting that matters is how much healthy data the reference sees. With only 1 hour of baseline, every method's error roughly doubles, with tooth at 0.0039 and Isolation Forest at 0.0067. From 2 hours onward it is flat between 0.00179 and 0.00181. Adding healthy gears helps the learned models slightly, while helping the tooth-CI Mahalanobis model least, as outlined in Table B3.

Table B3. *Mean gear error against the number of training gears for the healthy model*

Method	1 training gear	3 training gears	Change
Tooth-CI Mahalanobis	0.00172	0.00160	−7 %
Isolation Forest	0.00417	0.00372	−11 %
All-CI Mahalanobis	0.00520	0.00493	−5 %

This is consistent with each gear being referenced to itself in the preprocessing. The pooled model only has to learn what normal deviation looks like, rather than each gear's absolute signature.

Several ablations of the tooth HI leave T1 exactly unchanged, including dropping FM4, M6A, or M8A, removing condition correction, or removing the steady filter. We checked that this is not a bug. The HI values do change, but their ordering across the 32 run ends does not, maintaining a Spearman rank correlation of 1.00. The pooled isotonic metric depends only on that ordering. The metric is therefore blind to these changes, and NA4 is the only tooth CI whose removal changes the ranking, adding 0.0004. Isolation Forest is the least stable method.

- Random seed: Five seeds give T1 scores ranging from 0.0026 to 0.0040 with a standard deviation of 0.00054. That spread represents more than a third of the gap between Isolation Forest and the best model.
- Trees: Increasing tree count reduces error steadily from 0.0035 at 50 trees down to 0.0028 at 500 trees, which is consistent with seed noise.
- Max features of 0.25 to 1.0 yield 0.0019 to 0.0032. Max samples of 64 to 1024 yield 0.0021 to 0.0032.

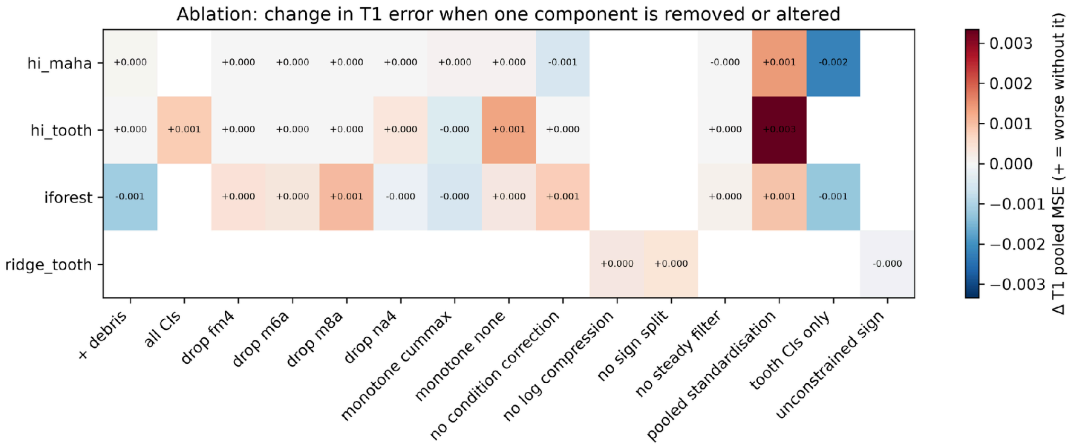


Figure B6. *Ablation: change in T1 when one component is removed or altered (red = worse without it). Rows: all-CI Mahalanobis, tooth HI, Isolation Forest, and the supervised ridge model from Part A, whose three cells are discussed in Part A's Feature analysis.*

Discussion

Unsupervised learning Part B

We expected the more flexible models like an Isolation Forest to beat a mean of four clipped z-scores. They did not. The no-model tooth HI is statistically tied with the best learned model, and every model fit on all 12 CIs lost to it. The dominant source of novelty in this rig is not damage but reassembly, as each stop and restart shifts the global vibration signature by more than an early spall does. A novelty detector finds the largest departure from normal, and here that departure is not the one we care about. A second surprise was that a label-free selection rule in Table B2 chose the same model as the labels.

Challenges and responses.

- Restart shifts dominated the all-CI scores. We responded with physics-motivated feature selection focusing on the tooth-local CIs and a directional score where only upward departures count, rather than increasing model capacity.
- Experiments only exist for four gears. We evaluated at the gear level, refused minute-level splits, reported per-gear results next to every pooled number, and stated the $1/2^4$ p-value floor plainly instead of quoting a misleading CI over 10,000 correlated minutes.
- Tuning on labels would make the method no longer unsupervised. We fixed the settings in advance, used label-free fitness for selection, and quantified the cost of label tuning separately in Part A.

Proposed Future Improvements

- Restart-Aware Baseline: Re-reference each run to its own first healthy minutes or model the restart shift explicitly as a change point. This would allow all-CI models to contribute the information in their extra features.
- Use high-frequency vibration waveforms to learn representations directly, for example with an autoencoder trained on healthy minutes. Its reconstruction error would replace hand-crafted CIs and be evaluated under the same LOGO protocol.
- Cluster run-level trajectories across more gears, to tell gradual-wear gears (B, C, F) from sudden-spall gears (A). The first group gives early warning and the second does not.

Ethical Considerations

- Asymmetric Costs: A missed spall on a rotorcraft or wind-turbine gearbox can cost lives or a machine, whereas a false alarm costs an inspection. The tooth-CI models give no warning on Exp-A, whose spall arrives within a single run. The models that flag Exp-A earlier do so by alarming at a rig restart, and the all-CI ridge trades fewer false alarms for a missed spall. We report caught and missed events alongside lead times for every accuracy number so readers can weigh both error types, and we would not deploy any of these models as the sole trigger for maintenance.
- Unsupervised Alarm Trust: An unsupervised HI raises an alarm when a gear departs from its healthy past, so main risks come from over-trust and misinterpreting its meaning. On Gear A the spall appeared within a single run and no tooth-CI HI warned prior to it. Operators must not treat no alarm as no damage. With a 95th-percentile threshold, about 5 percent of healthy minutes alarm by design, and restarts trigger more. Frequent false alarms cause alarm fatigue, so we report false-alarm counts next to every accuracy figure and would require a persistence rule before acting on an alarm.
- Automation complacency. A health index that is usually right invites operators to skip inspections. The HI is meant to schedule inspections, not replace them, and the report says so.

Statement of Work

Team member	Contribution
Garrett Kohler	Data extract and HDF5 flattening; exploratory analysis and photo proxy; unsupervised health index and submission pipeline, ethical considerations, similar work, future work, unsupervised learning discussion;
Jason Rushton	Supervised learning models, EDA, Feature engineering, data pre-processing for Supervised model optimization, Report writing for Part A(Supervised Learning), Creating standard for manual scoring of gears
Sergio Lopez	XGBoost supervised regression models; vibration feature validation; model performance evaluation and visualization; manual labeling and scoring of gear tooth images to support model training and validation.

Github: https://github.com/jrush331/Gear_Spall_Damage_Prognostics

Appendix A. Bibliography

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Appendix B. Data schema

Per-second table (`data/<split>/lf_1s.parquet`, one row per gear, run and second; 47 signal columns listed in `data/channels.csv` plus `experiment`, `run_num`, `hours`):

Group	Columns	Examples	Used as
Condition indicators	26 (`ci_*`)	`ci_fm4, ci_m6a, ci_m8a, ci_na4, ci_e_ratio, ci_kurtosis_tsa, ci_rms_tsa, ci_alr, ci_fm0, ci_crest_tsa, ci_kurtosis_e_op, ci_rms_e_op (the 12 Part B inputs); ci_*star, ci_fleet_wtmade (Part A candidate features, as *_norm and distance-from-running-minimum transforms); ci_rms, ci_ptp, ci_v_peak_mmps (unused)`	Model inputs after log, condition correction and per-gear standardisation
Context	5 (`ctx_*`)	`ctx_pau_speed, ctx_pau_torque, ctx_temperature, ctx_accel1_rms, ctx_accel2_rms`	Steady-state filter (speed, torque); regressors (temperature, torque)
Oil debris	16	`dm4500_bin_1..8, icm2_bin_1..8`	Debris baseline (`dm4500_bin_2..7`) and one ablation

Labels (`data/labels/condition_labels.csv`): `experiment` (Exp-A/B/C/F), `tooth` (1–28; 1–14 for Exp-C), `run_num` (inspection after this run), `score` (1 healthy … 4 severe spalling, fractional allowed). Derived per run end: `mean_ex` = mean(score − 1) (T1), `max` = worst tooth. `condition_labels_adjusted.csv` is a later revision (896 rows, all 28 teeth of Exp-C) evaluated separately in `models/eval/out_adjusted/`. `spall_scores.csv` is the earlier Exp-A/B-only file used in Table A1.

Run ends (`models/eval/out/runs.csv`): one row per approach and run end with `score` (monotone), `causal` (running max), `threshold`, `photo` (T2), `mean_ex` (T1), `max` and the `spall` flag.

Appendix C. Script and notebook catalog

Path	Purpose
`scripts/phm_data.py`	Download and extract the challenge data into one flat directory per split with a provenance manifest
`scripts/flatten_hdf5.py`	Flatten per-minute HDF5 files into `lf_1s.parquet` and `particles.parquet` per split
`eda/scripts/build_min.py`, `compute.py`, `novelty.py`	Minute tables, summary statistics, correlation and early novelty scores for the EDA notebook (`eda/out/data.json`)
`eda/scripts/photo_proxy.py`, `photo_proxy2.py`	By-eye photo damage proxy (T2) from the tooth photographs
`eda/spall_figures.ipynb`, `spall_figures_plotly.ipynb`	Milestone I exploratory figures
`models/spall_hi.py`	Preprocessing pipeline, healthy-state models, run-end HI, label-free selection and challenge submissions
`models/spall_supervised.py`	Non-negative ridge on labels; 96-setting label-tuned search; `labelled_*` submissions
`models/eval/harness.py`	Dataset, LOGO folds, T1/T2 metrics, bootstrap, sign-flip test, causal detection and lead time
`models/eval/adapters.py`	One adapter per approach: six HIs, Isolation Forest, ridge (tooth/all), label-tuned (nested/non-nested), hours, debris
`models/eval/compare.py`	Table A2/A3, winner rule, trajectories, per-gear dots, lead-time figure
`models/eval/ablation.py`, `sensitivity.py`, `curves.py`	Table A4/Figure B6 and Table A5, Figure A3
`models/eval/failures.py`	Residual ranking, hypothesis confirmation, exact/SHAP/LIME attributions, seed study (Tables A6–A7)
`models/eval/fig_timeline.py`, `fig_ridge_coefs.py`	Figure 1 and Figure A2
`models/eval/fig_early_warning.py`, `fig_model_vs_hours.py`, `fig_unsupervised.py`	Part B figures
`models/eval/run_all.py`	Regenerates every output in `models/eval/out/`
`models/eval/build_notebook.py`, `cells.py`, `spall_eval.ipynb`	Report notebook that reads `out/` only
`tests/test_spall_hi.py`, `tests/test_eval_harness.py`	37 tests: pipeline invariants, fold hygiene, metric values, adapters, winner rule, attributions, end-to-end `run_all`
`docs/report/check_numbers.py`, `build_pdf.py`, `build_docx.py`	Verify the numbers in this document against the output tables; build the PDF (headless Chrome) and the `.docx`

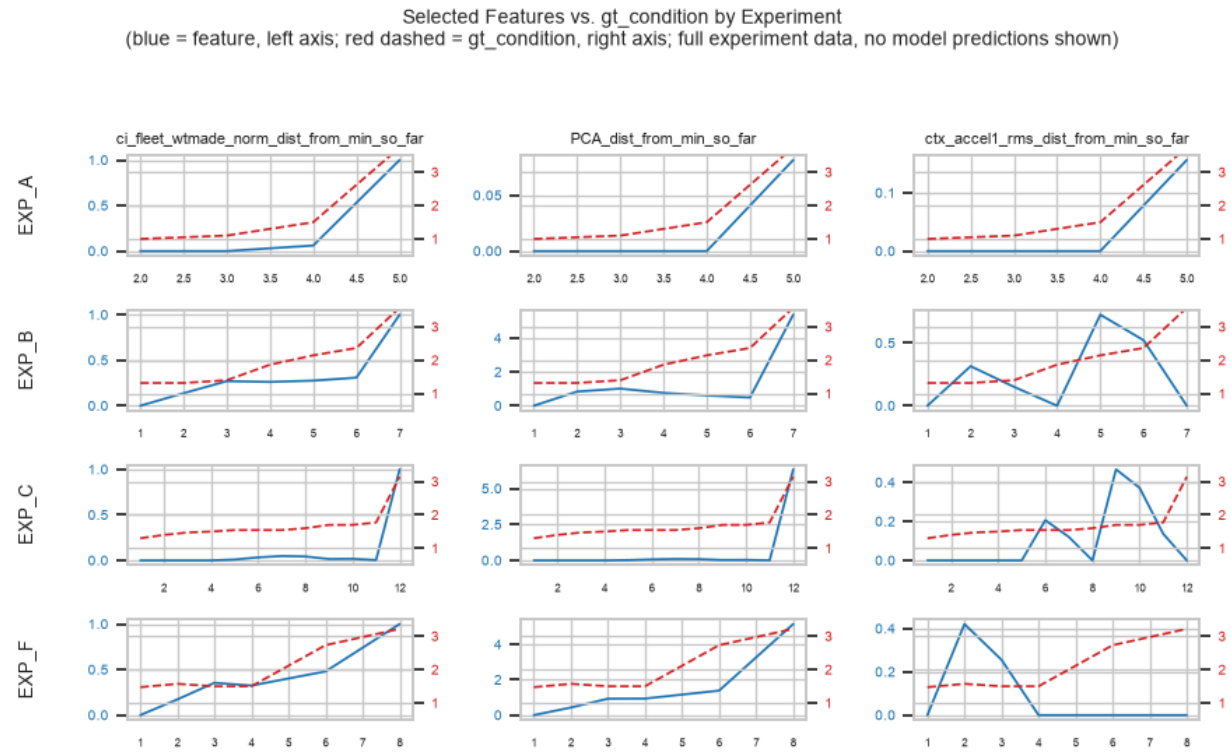
Appendix D. Reproducibility

```
# data (once; ~1 GB)
.venv/bin/python scripts/phm_data.py small && .venv/bin/python scripts/flatten_hdf5.py
# submissions and the original two-gear check
PYTHONPATH=models .venv/bin/python models/spall_hi.py
PYTHONPATH=models .venv/bin/python models/spall_supervised.py
# every table, figure and quoted number in this report (~30 s)
.venv/bin/python -m models.eval.run_all
.venv/bin/python -m models.eval.fig_unsupervised
# checks
.venv/bin/python -m pytest tests -m "not slow"
.venv/bin/python docs/report/check_numbers.py
```

Fixed parameters. Seed 696 everywhere (Isolation Forest, bootstrap). Steady state: 1400 ± 20 rpm, torque within 5 % of the gear median. Run-in skip 0.5 h; healthy window 0.5–6 h; run-end value = 90th percentile of the last 60 steady minutes; monotone step = isotonic regression (running max for alarms). Damage scale 3 (grade 4 − 1). Ridge: α = 10, `positive=True`, target `mean_ex`, sign split + log1p. Label-tuned grid: 4 tooth subsets × 2 forms × 2 windows × 3 tails × 2 monotone steps = 96. Alarm threshold: 95th percentile of the approach's score on the training gears' healthy minutes. Bootstrap: 2,000 resamples. Tie tolerance for the winner rule: 0.001 in T1. Failure-example confirmation: top-5 residual rank; seed study: 10 seeds.

Software. Python 3.12.13, scikit-learn 1.9.1, scipy 1.18.1, pandas 3.0.6, numpy 2.5.3, pyarrow 25.0.1, h5py 3.16.0, shap 0.53.0, lime 0.2.0.1, matplotlib 3.11.2, python-docx 1.2.0.

Appendix E. Feature Visualization Supervised Learning



Acknowledgment

The team used Claude Code to assist with portions of this project, including code development and editing of the report text; all analysis decisions, results, and interpretations are the authors' own and were verified by the team.